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HL7 Version 2 Implementation Guide: Clinical Genomics; Fully LOINC-Qualified Genetic Variation Model, Release 2

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1. Introduction

1.1 FORWARD

Publication of this Technical Report that has been registered with ANSI has been approved by the accredited standards developer Health Level Seven International (HL7), 3300 Washtenaw Ave., Suite 227, Ann Arbor MI 48104, USA. This document is registered as a Technical Report according to the procedures for the Registration of Technical Reports with ANSI. This document is not an American National Standard and the material contained herein is not normative in nature. Comments on the content of this document should be sent to the HL7 Clinical Genomics Work Group, 3300 Washtenaw Ave., Suite 227, Ann Arbor MI 48104-4261 or hq@hl7.org. This document was registered as a Technical Report because coded and structured clinical genetic test results are key for clinical decision support in the electronic health record, augmentation of patient data in clinical data warehouses, research studies, clinical trials, as well as other areas of electronic health data management.

1.2 OVERVIEW

The *HL7 Version 2.5.1 HL7 VERSION 2 IMPLEMENTATION GUIDE: CLINICAL GENOMICS; FULLY LOINC-QUALIFIED GENETIC VARIATION MODEL, RELEASE 2 (US Realm)* details structuring a genetic test results into the electronic health record utilizing HL7 version 2.5.1. This implementation guide is modeled after the *HL7 Version 2.5.1 Implementation Guide: Orders And Observations; Interoperable Laboratory Result Reporting To EHR (US Realm), Release 1,* and covers the reporting of genetic test results for sequencing and genotyping based tests where identified DNA variants are located within a gene. For greater understanding of the area of clinical genetic laboratory testing, the reader should refer to AHIC's *Personalized Healthcare Detailed Use Case (Available at: http://www.hhs.gov/healthit/usecases/phc.html)*.

In March, 2008, the United States Department of Health and Human Services, Office of the National Coordinator for Health IT published the *Personalized Healthcare Detailed Use Case* (Click here to see the use case) in response to a request and specifications from the American Health Information Community. The use case focuses on supporting secure access to electronic genetic laboratory results and interpretations for clinical care, as well as family history and associated risk assessments by authorized parties and is driven by the need for timely electronic access to ordered, referred and historical genetic laboratory results, and family history. Ordering clinicians receive genetic laboratory test results as a response to an order by having the genetic test results sent either directly to the clinician's EHR system (local or remote), or to another clinical data system, in support of the provisioning of historical results.

At the time of balloting Release 1 of this guide, two healthcare providers and a CLIA certified genetic testing laboratory were piloting the information model detailed in this implementation guide. At the time of writing Release 2, two additional CLIA certified laboratories and a genomic test development/research laboratory were added to the list of piloting organizations. See section 1.6 for details. Content within Release 2 was developed in response to feedback from these piloting organizations. See section 1.4 for email contacts for submission of suggestions of future extensions. The complexity of genetic data requires additional coding of the message components using LOINC. These codes are listed in tables in section 7. LOINC coding has several advantages including more robust representation of the data when persisted in a database, increased accuracy when supporting multiple HL7 message formats, and consistency of representation for clinical decision support. At the time of writing Release 2, the Clinical Genomics Work Group has been writing an implementation guide detailing how these codes might be used within an HL7 v3 CDA based message. This project is entitled: *Implementation Guide for CDA Release 2 Genetic Testing Report (GTR)*.

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The chapters in this guide that describe messaging infrastructure, abstract message syntax, and segment and field descriptions are based on chapters from the parent implementation guide entitled **HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007**. This guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required).

1.3 RELEASE 2 NOTES

Release 1 of the HL7 v2.5.1 Implementation Guide: HL7 VERSION 2 IMPLEMENTATION GUIDE: CLINICAL GENOMICS; FULLY LOINC-QUALIFIED GENETIC VARIATION MODEL has been extended in Release 2 to support codified transmission of:

1) Large Genetic Profiling Result Sets (using genotyping technology) with no encoded interpretation within the results - but linkage to external knowledge resources,
2) Details of Commercially Available Gene Chips, and
3) Genetic Test Definition.

Release 2 follows the same model as Release 1, by continuing to be modeled after the *HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007. This guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required). As such, chapters in this guide that describe messaging infrastructure, abstract message syntax, and segment and field descriptions are based on chapters from the parent implementation guide*.

1.4 PURPOSE

The *HL7 VERSION 2 IMPLEMENTATION GUIDE: CLINICAL GENOMICS; FULLY LOINC-QUALIFIED GENETIC VARIATION MODEL (RELEASE 1 & 2) (US Realm)* is modeled after established laboratory reporting standards for genetic test results for sequencing and genotyping based tests where identified DNA sequence variants are located within a gene. This includes testing for DNA sequence variants that are associated with a disease (or risk for developing the disease) and pharmacogenomic applications, such as predicting a patient's responsiveness to drug therapy and drug metabolism rate, based on DNA sequence variants associated with these drug responses. It should be noted that genetics (both inherited, germline DNA variants and acquired, somatic DNA variants) is only one component in determining patient clinical state. Other contributions include patient and family health history, tissue pathology findings, diet, medications, and behavioral and environmental variables.

1.5 AUDIENCE

This guide is designed to be used by analysts and developers who require guidance on the reporting of genetic test results generated through gene or partial gene sequencing or genotyping clinical diagnostic tests. Users of this guide must be familiar with the details of HL7 message construction and processing. This guide is not intended to be a tutorial on that subject.

1.6 SCOPE

This guide covers the reporting of DNA based genetic test results performed using sequencing or genotyping technology for the identification of DNA sequence variations contained within a gene. This includes testing for DNA variants associated with disease or pharmacogenomic response to drugs (efficacy or metabolism).

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- **Use of Vocabulary Standards** This guide calls for specific vocabulary standards for the exchange of laboratory information. Use of standard vocabularies is important for a number of reasons. Use of standard vocabularies allows broad distribution of healthcare information without the need for individual institutions to exchange master files for data, such as test codes, result codes, etc. Each institution maps its own local vocabularies to the standard code, allowing information to be shared broadly, rather than remaining isolated as a single island of information. Standard vocabularies, particularly coded laboratory results, enable more automated decision support for patient healthcare, as well as more automated public health surveillance of populations. In addition, the codified nature of these findings supports the use of these vocabulary standards in other messaging types, and provides a starting place for representation of this data within the database/knowledgebase.
- **Extension of Vocabulary Standards** We would like to know of challenges implementers may have in using this implementation guide and set of vocabularies. We would also like to learn of suggested extensions and/or modifications or new scenarios/use cases which should be supported. To submit suggestions, please send them to the HL7 Clinical Genomics listserve, or contact Mollie Ullman-Cullere (mullmancullere@partners.org) or Grant Wood (grant.wood@imail.org).

1.7 ASSUMPTIONS

Assumptions are summarized as follows:

- Infrastructure is in place to allow accurate information exchange between information systems.
- Providers access laboratory test results through either an EHR or a clinical data system.
- Privacy and security has been implemented at an acceptable level.
- Participants agree to all standards, methodologies, consent, privacy and security.
- Legal and governance issues regarding data access authorizations, data ownership and data use are outside the scope of this document.
- The order, paper or electronic, associated with the laboratory result, contains sufficient information for the laboratory to construct the laboratory result message properly.

1.8 CONVENTIONS

The following conventions have been used in establishing this guide:

- The rules outlined in *HL7 2.5.1*, *Chapter 2*, *Section 2.12*, *Conformance Using Message Profiles*, were used to document the use case for, and constraints applied to, the messages described in this guide.
- Data types have been described separately from the fields that use the data types. For details regarding data type field lengths, please refer to *Section 2.1.3*, *Lengths* in this document.

1.9 PILOT PROJECTS

This information model is based on the HL7 Version 3 Genetic Variation model. A message consistent with this model has been piloted for 5+ years transmitting genetic test results between the Laboratory for Molecular Medicine, Partners HealthCare Center for Personalized Genetic Medicine (formerly the Harvard – Partners Center for Genetics and Genomics) and Partners Healthcare's electronic medical record. For the purpose of this work, the model was translated from HL7 Version 3 to HL7 Version 2.5.1. In addition, within Release 1, the model has been extended to reflect lessons learned. This includes association of findings to SNOMED coded disease or RxNORM coded medications. Within Release 2, the model was further extended to reflect lessons learned from additional pilots. The information model detailed within this implementation guide has been piloted by the following organizations:

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1) Genetic Testing Laboratories
Laboratory for Molecular Medicine, Partners HealthCare Center for Personalized Genetic Medicine (formerly Harvard – Partners Center for Genetics and Genomics), Cambridge, MA
ARUP, University of Utah, Salt Lake City UT
Center for Advanced Molecular Diagnostics, Brigham and Women's Hospital, Boston MA
Center for Cancer Genomic Discovery, Dana-Farber Cancer Institute, Boston MA

2) Receiving Provider Electronic Medical Records
Partners Healthcare, Boston, MA
Intermountain Healthcare, Salt Lake City, UT

3) Systems for Discovery Research and Translational Medicine, including results viewer, research data warehouse, and analytics
Dana-Farber Cancer Institute, Boston MA

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2. Messaging Infrastructure

The V2 Genetic Variation model uses the same messaging infrastructure as described in Chapter 2, Page 3 of the parent implementation guide entitled **HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007**. The guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required).

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3. Message Profile – Genetic Laboratory to EHR

3.1 USE CASE MODEL

TABLE 3-1 – USE CASE LABORATORY TO EHR	
Description	The *Personalized Healthcare Detailed Use Case* was published by the Office of the National Coordinator for Health Information Technology (ONC). This document focuses on the subset of the use case that applies to the exchange of laboratory results between the genetic testing laboratory and the EHR. The guide covers genetic test results for sequencing and genotyping based tests where identified variants are located within a gene. This includes testing for DNA variants that are associated with a disease (or risk for developing the disease) and pharmacogenomic applications, such as predicting a patient's responsiveness to drug therapy and drug metabolism rate, based on DNA variants associated with these drug responses. It should be noted that genetics (both inherited germline DNA variants and acquired somatic DNA variants) is only one component in determining patient clinical state. It does not cover querying patient demographics or laboratory results. It does include acknowledgments of receipt of transactions. The complexity of genetic data requires additional coding of the message components using LOINC. These codes are listed in tables in section 7.4.
Actors	Laboratory Result Sender – The laboratory result sender actor is an application capable of performing laboratory testing on specimens. The laboratory application is capable of transmitting the results of laboratory testing to a receiver. In the use case, the laboratory result sender is identified as a "Laboratory Organization." Laboratory Result Receiver – The laboratory result receiver is an application capable of receiving results of laboratory testing. Typically this actor represents an EHR application. The laboratory result receiver may be associated with the ordering provider or another provider, commonly referred to as a "copy-to provider," that needs to have access to the results. In the use case, the laboratory result receiver is identified as either the "Clinician" or "Data Repository."
Assumptions	Assumptions are summarized as follows: Infrastructure is in place to allow correct information exchange between information systems. Providers access laboratory test results either through an EHR or a clinical data system. Privacy and security has been implemented at an acceptable level. All participants agree to all standards, methodologies, consent, privacy and security. Legal and governance issues regarding data access authorizations, data ownership and data use are outside the scope of this document. The following are preconditions[1] for the use of this profile: The order contains the unambiguous names and electronic addresses for the other authorizedproviders of care. When needed, the patient is registered in a Patient ID Cross-Referencing system that includes both

[1] From *HITSP Interoperability Specification: Send Laboratory Result Message to Ordering Clinician and Providers of Care Transaction Package*, dated

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	the laboratory patient ID and the clinician's patient ID. For the electronic laboratory result, the laboratory has transformed any local codes into HITSP-specified terminologies before transmission. Additional Preconditions: A valid order for laboratory testing exists.

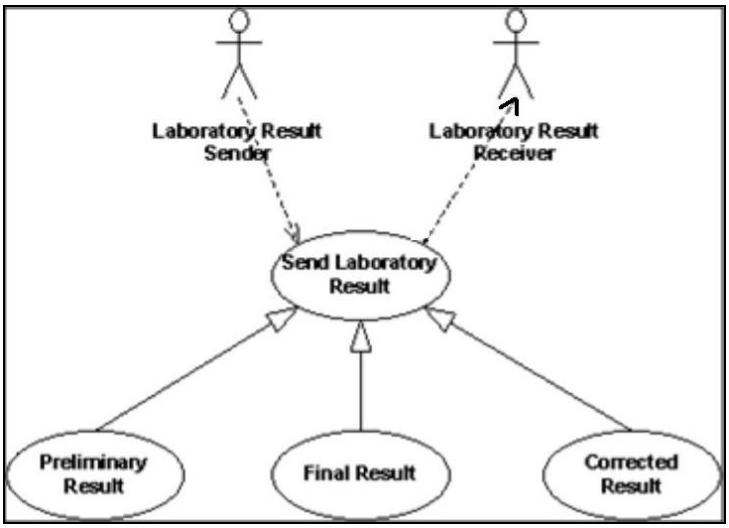

Figure 3-1. Send Genetic Laboratory Result Use Case Model

3.2 DYNAMIC INTERACTION MODEL

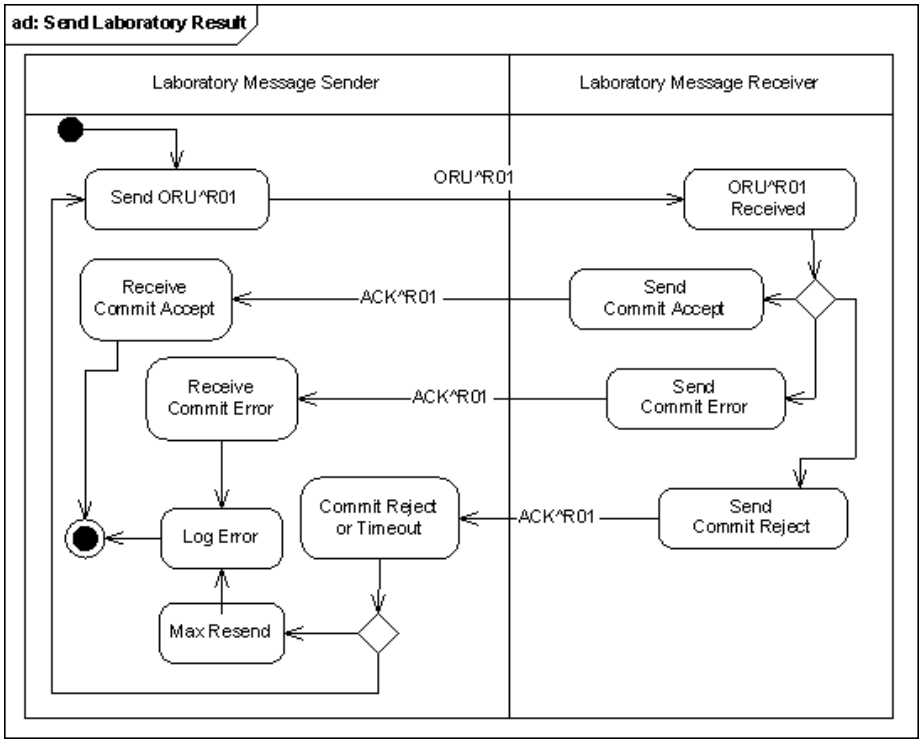

Figure 3-2. Activity Diagram for Send Genetic Laboratory Result Use Case

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3.3 DYNAMIC DEFINITION

TABLE 3-2 – DYNAMIC DEFINITION	
Item	**Value**
Profile ID	USLabReport
HL7 Version	2.5.1
Accept Acknowledgement	AL – Always
Application Acknowledgement	For valid values, refer to HL7 Table 0155 – Accept/Application Acknowledgment conditions in section 5.2.1 in HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007. The guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required). .
Acknowledgement Mode	Immediate
Profile Type	Realm Constrainable Profile
Message Types	ORU^R01^ORU_R01, ACK^R01^ACK
Encoding	ER7 (required) 2.5.1 XML (optional)

3.4 INTERACTIONS

TABLE 3-3 – INTERACTIONS							
Event	**Description**	**Usage**	**When Used**	**Message Type**	**Receiver Action**	**Sender**	**Data Values**
Order Received, No specimen	Order received; specimen not yet received	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=O
Specimen Received	No results available; specimen received, procedure incomplete	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=I
Procedure Scheduled	No results available; procedure scheduled, but not done	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=S
Preliminary Result	Preliminary: A verified result is available, final results not yet	R	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=P

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TABLE 3-3 – INTERACTIONS							
Event	Description	Usage	When Used	Message Type	Receiver Action	Sender	Data Values
	obtained						
Partial Result	Some, but not all, results available	O	Some Final Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=A
Unverified Result	Results stored; not yet verified	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=R
Final Result	Final results; results stored and verified. Can only be changed with a corrected result.	R	Final Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=F
Correction	Correction to results	R	Corrected Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=C
Testing Not Done	No results available; Order canceled.	O	Cancelled Test	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=X
No Order	No order on record for this test. (Used only on queries)	-	-	varies	NA	Laboratory Result Sender	ORC-1=RE OBR-25=Y
No Patient Record	No record of this patient. (Used only on queries)	-	-	varies	NA	Laboratory Result Sender	ORC-1=RE OBR-25=Z
Commit Accept	Enhanced mode: Accept acknowledgment : Commit Accept	R	All Cases	ACK^R01^ ACK	None	Laboratory Result Receiver	MSA-1=CA
Commit Error	Enhanced mode: Accept acknowledgment : Commit Error	R	All Cases	ACK^R01^ ACK	None	Laboratory Result Receiver	MSA-1=CE
Commit Reject	Enhanced mode: Accept acknowledgment : Commit Reject	R	All Cases	ACK^R01^ ACK	None	Laboratory Result Receiver	MSA-1=CR

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4. Messages

The V2 Genetic Variation model uses the same messages as described in Chapter 4, Page 29 of the parent implementation guide entitled **HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007**. The guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required).

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5. Segment and Field Descriptions

The V2 Genetic Variation model uses the same segment and field descriptions as described in Chapter 5, Page 35 of the parent implementation guide entitled **HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007**. The guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required).

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6. Nomenclatures, Code Systems and Value Sets

6.1 VOCABULARY CONSTRAINTS

6.1.1 Genetic Tests, Testing Context, Interpretation Code, and Genetic Data

6.1.1.0 LOINC

TABLE 6-1 – LABORATORY LOINC	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	All LOINC laboratory result codes
Minimum attributes of the component	HL7 value sets not established. Considered value sets should include: HEDIS (Health plan Employer Data and Information Set) reported tests accounting for 95% of routine laboratory orders. Proposed value sets for micro and cytology codes per HITSP/C80. Category A, B, & C bioterrorism agents/diseases. Public Health jurisdiction and Federal reportable disease conditions. The LOINC codes for reporting clinical genomics variation are listed in this implementation guide explicitly – and were developed and tested as described below.
Other Comments	LOINC - Vocabularies and code sets, useful in the reporting of genetic test result data into the EHR, in formats that can be leveraged by clinical decision support, have been defined as a result of the 2 year clinical pilot of the HL7 Version 3 Genetic Variation model. These vocabularies and code sets have be submitted to LOINC, and through ongoing collaborations between the National Library of Medicine's Lister Hill Center for Biomedical Communication, Partners HealthCare Center for Personalized Genetic Medicine (formerly the Harvard – Partners Center for Genetics and Genomics), Partners Healthcare, and Intermountain Healthcare, these vocabularies and codes will be piloted more broadly. In addition, the above collaborators have detailed these vocabularies and code sets in the HL7 implementation guide, balloted in Fall 2009, entitled: HL7 Version 2 Implementation Guide: Clinical Genomics; Fully LOINC-Qualified Genetic Variation Model, Release 1. The full LOINC data base can be downloaded from the Regenstrief Institute, Indianapolis, Indiana at http://loinc.org/.

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6.1.2 Associated Disease and/or Drug

6.1.2.0 SNOMED-CT

TABLE 6-2 - SNOMED-CT	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	SNOMED-CT
Minimum attributes of the component	SNOMED-CT FDA SPL Problem List Subset
Other Comments	FDA SPL Problem List Subset available at http://www.fda.gov/ForIndustry/DataStandards/StructuredProductLabeling/ucm163377.htm . The SNOMED terminology is used in the coding of disease associated with sequence variants or genes. Utilization of SNOMED provides linkage of genetic data with other clinical data stored in clinical applications.

6.1.2.1 RxNORM

TABLE 6-3 - RXNORM	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	RxNORM
Minimum attributes of the component	Medication List Subset
Other Comments	Use RxNORM ingredient codes to identify drugs that are the target of pharmacogenomics studies. Utilization of RxNORM provides linkage of genetic data to other clinical data stored in clinical applications. An overview of RXNorm can be obtained from http://www.nlm.nih.gov/research/umls/rxnorm/overview.html

6.1.3 Genes

6.1.3.0 **HGNC** gene symbols (required)

TABLE 6-4 - HGNC	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	HGNC
OID	2.16.840.1.113883.6.281
Minimum attributes of the component	Gene symbol
Other Comments	Human Gene Nomenclature Committee (HGNC maintains a database of gene names and symbols. They are a non-profit body which is jointly funded by the US National Human Genome Research Institute (NHGRI) and the

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	Wellcome Trust (UK). They operate under the auspices of Human Genome Organization. The database can be found at http://www.genenames.org/.

6.1.4 Sequence Variations

6.1.4.0 **HGVS** (required)

TABLE 6-5 – HGVS	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	HGVS
OID	2.16.840.1.113883.6.282
Minimum attributes of the component	Sequence variation
Other Comments	Human Genome Variation Society (HGVS) Nomenclature standards for the description of sequence variations are maintained at http://www.hgvs.org/mutnomen/. This standard is well accepted by the clinical genetic community and is extended on an ongoing basis to support genetic findings. HGVS maintains a tool for the generation/translation/verification of HGVS nomenclature. This tool can be found at: http://www.mutalyzer.nl/2.0/index

6.1.4.1 **dbSNP** (optional, but highly recommended)

TABLE 6-6 – DBSNP	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	dbSNP
OID	2.16.840.1.113883.6.284
Minimum attributes of the component	Rs number and nucleotide change
Other Comments	The Single Nucleotide Polymorphism database (dbSNP). Is maintained by National Center for Biotechnology Information. Available at: http://www.ncbi.nlm.nih.gov/projects/SNP/ Databases and knowledgebases defining sequence variants will be increasingly important. Although sequencing based tests which can result in the identification of novel variants require HGVS nomenclature standards for complete results reporting, genotyping tests which probe for the existence of known variants can additionally report results using an 'RS number' (i.e. identifier in dbSNP) and the associated nucleotide change. (Within the clinical environment results reporting using HGVS nomenclature is required with an option to additionally specify the RS number.)

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6.1.4.2 COSMIC (optional)

Variants/Mutations can also be reported with a COSMIC mutation identifier associating the findings with internationally compiled cancer mutation data.

TABLE 6-10 - COSMIC	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	COSMIC (Catalogue Of Somatic Mutations In Cancer)
Responsible Body	Sanger Institute
OID	2.16.840.1.113883.3.912
Minimum attributes of the component	COSMIC ID
Other Comments	Catalogue Of Somatic Mutations In Cancer (COSMIC) serves as a repository for somatic mutations identified in specific cancer specimens. These mutations are recorded associated with structured description of the specimen. Available at: http://www.sanger.ac.uk/genetics/CGP/cosmic/

6.1.5 Reference Sequences (required)

Reference sequences are the baseline from which variation is reported. For example, sequence variants are identified in a patient by comparing the patient's DNA sequence to a reference sequence standard, used in the laboratory. Typically, differences between the patient and reference sequence are called sequence variation and are cataloged, interpreted and reported. Documentation of the reference sequence used is becoming increasingly important for normalization of results between laboratories. To meet this need NCBI is cataloging reference sequences used in clinical testing in the Core Nucleotide Database and can be referred to through the RefSeq identifiers. In collaboration with NCBI, the European BioInformatics Institute (EBI) is also developing a database of reference sequences called Locus Reference Genomic Sequences (LRG). The standard is still in draft status. Importantly, NCBI's RefSeq and EBI's LRG will contain the same reference sequences, annotations and cross references to each other.

6.1.5.0 RefSeq

TABLE 6-7 - REFSEQ	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	RefSeq
OID	2.16.840.1.113883.6.280
Minimum attributes of the component	RefSeq ID
Other Comments	National Center for Biotechnology Information (NCBI) Reference Sequences contained in Core Nucleotide database. (Note version numbers are required to uniquely identify the reference.) Available at: http://www.ncbi.nlm.nih.gov/nuccore?db=nuccore

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6.1.5.1 LRG

TABLE 6-8 - LRG	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	LRG
OID	2.16.840.1.113883.6.283
Minimum attributes of the component	LRG ID
Other Comments	Locus Reference Genomic Sequences an emerging standard led by the European Bioinformatics Institute. Available at: http://www.ebi.ac.uk/ebisearch/search.ebi?db=lrg&t=gene And http://www.lrg-sequence.org/page.php

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6.1.6 Publicly Available References (valuable for clinical and translational genomics)

6.1.6.0 OMIM (optional)

Clinical genetic/genomic results can be reported with an OMIM id for association to relevant information in the OMIM knowledgebase, which contains a compendium of information on genetic based disease, genes and mutations.

TABLE 6-9 – OMIM	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	OMIM (Online Mendelian Inheritance in Man)
Responsible Body	Johns Hopkins
OID	2.16.840.1.113883.6.174
Minimum attributes of the component	OMIM ID
Other Comments	Knowledgebase for genes, variants/mutations and genetic based phenotypes. Note this information includes somatic or acquired variants/mutations and phenotypes and is not limited to inherited variants/mutations and phenotypes. Available at: http://www.omim.org/ and through NCBI at http://www.ncbi.nlm.nih.gov/omim Additionally, dbSNP contains links to variants in OMIM.

6.1.6.1 PubMed (optional)

Coding of references may include PubMed ids to peer reviewed literature (e.g. publications within a medical journal).

TABLE 6-10 – PUBMED	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	PubMed
Responsible Body	United States National Library of Medicine
OID	2.16.840.1.113883.13.191
Minimum attributes of the component	PubMed ID
Other Comments	"PubMed comprises more than 20 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full-text content from PubMed Central and publisher web sites." Available at: http://www.ncbi.nlm.nih.gov/pubmed/

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6.1.6.2 PharmGKB (optional)

PharmGKB ids to community curated information on emerging pharmacogenomic associations.

TABLE 6-10 - PHARMGKB	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	PharmGKB (Pharmacogenomic Knowledge Base)
Responsible Body	Stanford University, Department of Genetics
OID	2.16.840.1.113883.3.913
Minimum attributes of the component	PharmGKB ID
Other Comments	The mission of PharmGKB is "to collect, encode, and disseminate knowledge about the impact of human genetic variations on drug response. We curate primary genotype and phenotype data, annotate gene variants and gene-drug-disease relationships via literature review, and summarize important PGx genes and drug pathways." Available at: http://www.pharmgkb.org/

6.1.6.3 ClinicalTrials.gov (optional)

ClinicalTrials.gov id maybe transmitted as part of the interpretation indicating which clinical trials the patient may qualify.

TABLE 6-11 - CLINICALTRIALS.GOV	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained:	ClinicalTrials.gov
Responsible Body:	U.S. National Institutes of Health and Lister Hill National Center for Biomedical Communications
OID	2.16.840.1.113883.3.1077
Minimum attributes of the component:	ClinicalTrials.gov Identifier
Other Comments:	"ClinicalTrials.gov is a registry of federally and privately supported clinical trials conducted in the United States and around the world. ClinicalTrials.gov gives you information about a trial's purpose, who may participate, locations, and phone numbers for more details. This information should be used in conjunction with advice from health care professionals." http://clinicaltrials.gov/ct2/home

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7.Logical Message Types

7.1 INTRODUCTION AND STRATEGY

The Genetic Test Result Reporting message is defined by a set of four nested LOINC panels, which serve as templates for the messages. In general, LOINC panel definitions include one LOINC code to identify the whole panel and a set of LOINC codes for each child element of that panel. A child element can also be a LOINC panel, and such panels can repeat, to provide a structure that can accommodate many reporting patterns. For each such child element, the panel definition also includes its data type, units of measure, optionality and answer list, as applicable. The definitional information for the four panels used to report Genetics Test Result Reports is included in this guide. It can also be obtained in electronic form from the LOINC web site.

In a message, each new panel of observations begins with an OBR segment that carries the LOINC ID for that panel, and is followed by a series of OBX's each of which carry caries the LOINC ID (OBX-3), and the value (OBX-5) of a particular observation. In a message, the first panel is the master panel for the reporting of genetic analysis. The first child panel delivers an overall summary of the study results and includes options for reporting the traditional narrative report, the overall study impression, and a few other items. Depending on the study being reported, the summary panel may contain variables required to summarize a pharmacogenomics study, or those required to summarize the genetic findings associated with a disease or the risk of a disease (see table 7-2). Next comes the discrete results panel (table 7-5), which contains the detailed results pay load in a series of one or more "*DNA sequence analysis discrete sequence variation panels"* (see table 7-6). This last panel repeats as many times as needed to report all of the variations of interest.

The pattern of panels is shown in the following diagram:

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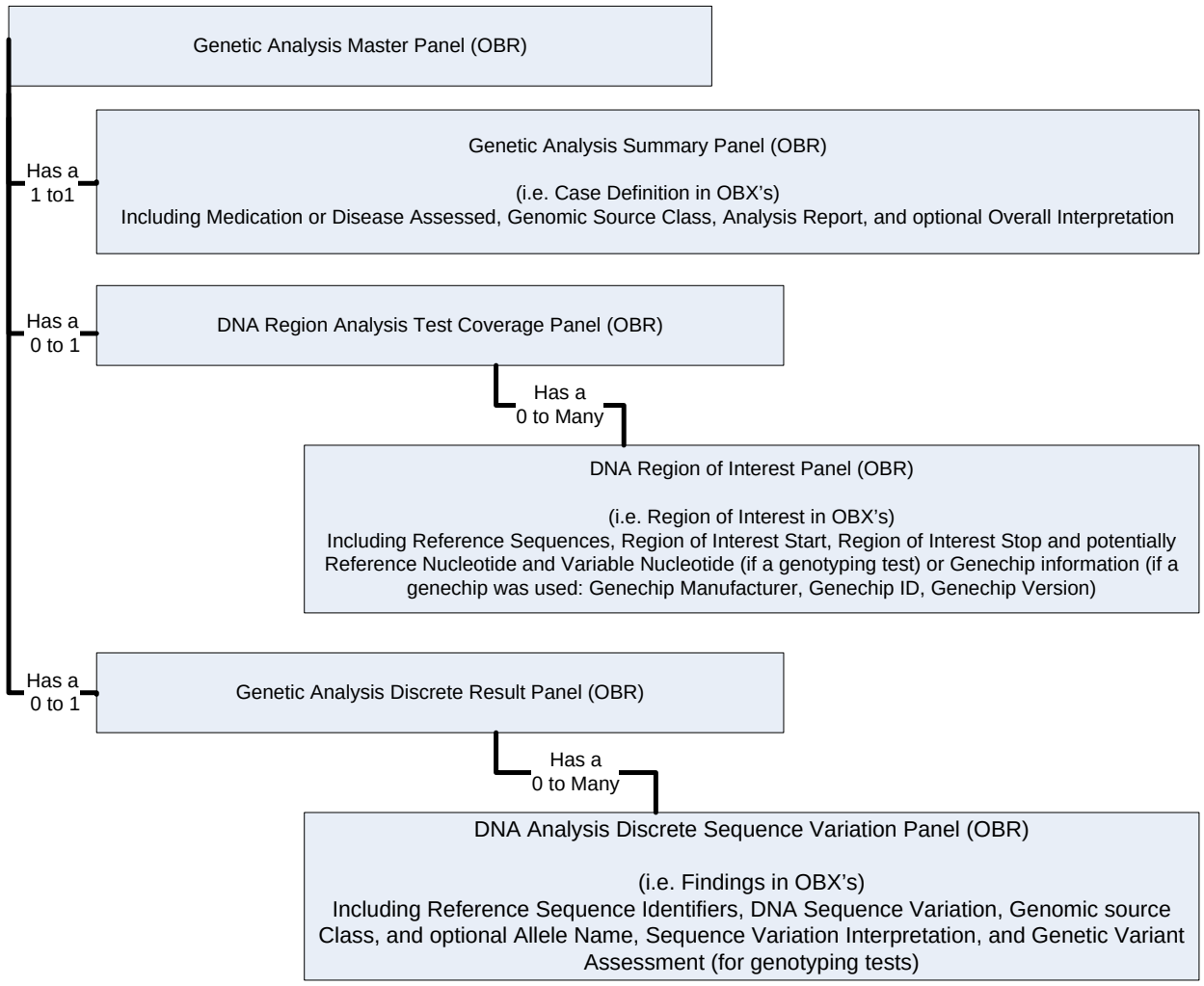

Figure 1. Object model of elements contained within the genetic results message. The master OBR (*Genetic Analysis Master Panel*) contains a child OBR, the *Genetic Analysis Summary Panel*. If information detailing the region of DNA examined is to be included in the HL7 message, then the *DNA Analysis Test Coverage Panel* will be added as a child OBR of the *Genetic Analysis Master Panel*. The *DNA Analysis Test Coverage Panel* itself has one or more child OBR's, the *DNA Region of Interest Panel*. If DNA sequence variations are identified, then the *Genetic Analysis Master Panel* will have another child OBR, the *Genetic Analysis Discrete Result Panel*. This child OBR (the *Genetic Analysis Discrete Result Panel*) itself has one or more child OBR's, the *DNA Analysis Discrete Sequence Variation Panel,* which are used to report genetic findings (sequence variations and gene alleles).

7.2 MESSAGE DEFINITIONS

The complexity of genetic data requires additional coding of the message components using LOINC. These codes are listed in tables 7.1 to 7.8. As already stated in the Introduction, LOINC coding has several advantages including more robust representation of the data when persisted in a database, increased accuracy when supporting multiple HL7 message formats, and consistency for clinical decision support, and applicability to many complex reporting requirements.

7.3 MESSAGE COMPONENTS

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The following tables 7-1 through 7-6 are NOT segment definitions. They specify the content of LOINC panels.

7.3.1 Test Interpretation

You can view the list of the individual test components included in each panel. The elements will be accompanied by a usage flag that will denote the expected appearance of the panel element in the panel when resulted. A usage flag is always one of three states:

- **R**equired. The panel element is always expected to be reported when the panel is resulted.

- **O**ptional. Report if available. The panel element may not be reported with a panel result depending upon institutional policy or capabilities of the reporting lab.

- **C**onditional. The panel element is a key finding in the panel report and should be assumed to be negative, absent or not present if the panel result does not include data for this element.

7.3.1.0 Genetic Analysis Master Panel

The *Genetic Analysis Master Panel* is an OBR which will contain a child OBR with summary information of the genetic analysis (i.e. *Genetic Analysis Summary Panel*). In addition, if genetic biomarkers were identified (e.g. a DNA Sequence Variant), then the *Genetic Analysis Master Panel* will have a second child OBR, the *Genetic Analysis Discrete Result Panel* (which in itself will have child OBR's for the Sequence Variants).

TABLE 7-1 – GENETIC ANALYSIS MASTER PANEL

OBR/ OBX	DT	Usage	Card-inality	Value Set	LOINC Code	LOINC Element Name	Description/Comments
OBR		R	1..n		55233-1	Genetic Analysis Master Panel	This is the parent OBR for the panel holding the summary of genetic analysis (i.e. Genetic Analysis Summary Panel).

7.3.1.1 Genetic Analysis Summary Panel

The *Genetic Analysis Summary Panel* is used to report the summary of the genetic analysis. This will fall into several categories: including disease risk/diagnosis, carrier testing, drug metabolism, and drug efficacy. It includes the genetic report, appropriate overall interpretation answer list, disease or drug assessed, and genomic source class.

TABLE 7-2 –GENETIC ANALYSIS SUMMARY PANEL

OBR/ OBX	OBX-2 Value Type	Usage*	Card-inality	Value Set	LOINC Code	LOINC Element Name	Description/Comments
OBR		R	1..n		55232-3	Genetic Analysis Summary	The summary panel for a genetic analysis for one or more laboratory tests (e.g. analysis for disease risk,

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						Panel	diagnosis or pharmacogenetics) on a single accession.
OBX	CWE	C1		SNOMED	51967-8	Genetic disease assessed	A coded disease which is associated with the region of DNA covered by the genetic test (recommend SNOMED).
OBX	CWE	C1		RxNORM	51963-7	Medication Assessed	A coded medication accessed in a pharmacogenic test (recommend RxNorm).
OBX	CWE	R			48002-0	Genomic Source Class	The genomic class of the specimen being analyzed: Germline for inherited genome, somatic for cancer genome (e.g. DNA from tumor cells), and prenatal for fetal genome. LOINC Answer List values can be seen in Table 7.8.
If the study is intended to assess disease risk or diagnosis based on genetic findings, then the *Genetic Disease Analysis Overall Interpretation* is used (see below).							
OBX	CWE	C2			51968-6	Genetic Disease Analysis Overall Interpre-tation	Interpretation of all identified DNA Sequence Variations along with any known clinical information for the benefit of aiding clinicians in understanding the results overall in either the context of diagnosis or increased risk of disease. LOINC Answer List values can be seen in Table 7.8.
If reporting a genetic test specifically performed for carrier testing, then the *Genetic Disease Analysis Overall Carrier Interpretation* (below) should replace the *Genetic Disease Analysis Overall Interpretation.*							
OBX	CWE	C2			53039-4	Genetic Disease Analysis Overall Carrier Interpre-tation	Carrier Identification interpretation of all identified DNA Sequence Variations along with any known clinical information for the benefit of aiding clinicians in understanding the results overall. LOINC Answer List values can be seen in Table 7.8.
If the study is intended to assess drug efficacy, include the following LOINC term in the report.							
OBX	CWE	C2			51964-5	Drug Efficacy Analysis Overall	Overall predicted phenotype for drug efficacy for all DNA Sequence Variations identified in a

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						Interpre-tation	single case. LOINC Answer List values can be seen in Table 7.8.
If the study is intended to assess the effect on metabolism, include the following LOINC term in the report.							
OBX	CWE	C2			51971-0	Drug metabolism analysis overall interpret-tation	Overall predicted phenotype for drug metabolism for all DNA Sequence Variations identified in a single case. LOINC Answer List values can be seen in Table 7.8.
OBX	FT	O	0..1		51969-4	Genetic analysis summary report	Narrative report in disease diagnostic-based format, which is used for pharmacogenomic reporting as well and disease risk or diagnosis. These reports currently follow the same formatting recommendations.
OBX	FT	O			53577-3	Reason for Study Additional Note	Descriptive text to further annotate the coded *Reason for Study* associated with an ordered test.

* R = Required; O = Optional; C1 = Conditional, either of the C1 values is required according to the following rules. C1 Rules – If the test is performed for diagnostic confirmation or identification of pre-clinical disease risk, then 'Genetic Disease Assessed', or if the genetic test is performed for pharmacogenomic usage, then 'Medication Assessed' is used. C2 = Conditional, only one of the values can be used according to the following rules: C2 Rules – If the test is performed for diagnostic confirmation or identification of pre-clinical disease risk, then 'Genetic Disease Analysis Overall Interpretation' is used. If the test is performed for identification of carrier status, then 'Genetic Disease Analysis Overall Carrier Interpretation' is used. If a pharmacogenomic test is performed for indication of drug efficacy, then 'Drug Efficacy Analysis Overall Interpretation' is used. If a pharmacogenomic test is performed for indication of drug metabolism, then 'Drug metabolism analysis overall interpretation' is used.

7.3.2 Test Definition

7.3.2.0 DNA Region Analysis Test Coverage Panel

The *DNA Region Analysis Test Coverage Panel* is a child OBR of the *Genetic Analysis Master Panel*, and will contain child OBR's defining the discrete regions of interest. Currently, these are gene (or partial gene) sequencing or genotyping in the *DNA Region of Interest Panel*.

Prior to the widespread adoption of whole genome sequencing, knowing which region of the genome was tested is often important for clinical interpretation of the results and determining need for follow-up testing (specifically for negative or inconclusive results). For instance, if only genetic regions most commonly found to have mutations associated with a specific disease were tested, the clinician may want to follow-up with a test which examines the remaining genetic regions currently known to be associated with the disease. Additionally, as new regions are discovered (to be associated with the specific disease), the clinician may want to order these newly available genetic tests. For genotyping tests, the reference and variable nucleotides would also be reported, if known. If the test is performed using a genechip, then

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information on the manufacturer, genechip and version should also be listed. This information should be captured within the *DNA Region of Interest Panel.*

Important note: For sequencing tests, receiving systems should not assume that they can derive the patient sequence by comparing the reference sequence and the reported mutations, because polymorphisms and other differences from the reference (which are thought to be non-clinically important at the time of reporting) may not have been reported. Derivation of the patient sequence will only be possible if ALL differences from the reference sequence are reported in the findings. In this scenario, using the 'Sequence Variation Interpretation' to prioritize variants displayed within the electronic medical record would provide a mechanism to not overwhelm the clinician with data on benign variants (not currently deemed important for clinical care).

TABLE 7-3 – DNA REGION ANALYSIS TEST COVERAGE PANEL

OBR/ OBX	DT	Usage	Card-inality	Value Set	LOINC Code	LOINC Element Name	Description/Comments
OBR		R	1..n	N/A	51956-1	DNA Region Analysis Test Coverage Panel	This is the parent OBR for genetic panels for regions of interest (e.g. *DNA Region of Interest Panel).*

7.3.2.1 DNA Region of Interest Panel

The *DNA Region of Interest Panel* describes the regions examined for variation or confirmation of matching to the reference sequence (e.g. wild-type or 'normal'). For sequencing based tests, one or more regions will be sequenced and the region of interest will fall within the sequenced region.

TABLE 7-4 – DNA REGION OF INTEREST PANEL

OBR/ OBX	OBX-2 Value Type	Usage*	Card-inality	Value Set	LOINC Code	LOINC Element Name	Description/Comments
OBR		R	1..n	N/A	53041-0	DNA Region of Interest Panel	A set of observations which comprise the definition of a contiguous sequence of nucleotides in the genome. This is intended for use in genetic test coverage panels in genetic test reporting.

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OBX	CWE	R	0…1		48013-7	Genomic reference sequence identifier	This field carries the ID for the genomic reference sequence. The genomic reference sequence is a contiguous stretch of chromosome DNA that spans all of the exons of the gene and includes transcribed and non transcribed stretches. For this ID, use either the NCBI genomic nucleotide RefSeq IDs with their version number (see: NCBI.NLM.NIH.Gov/RefSeq), or use the LRG identifiers (see: http://www.lrg-sequence.org/)
OBX	CWE	O	0…1		69545-2	Region of Interest Start	This is the nucleotide number starting the region which was examined for variants (or wild-type confirmation) during testing, and is to be used in conjunction with *Region of Interest Stop*.
OBX	CWE	O	0…1		69546-0	Region of Interest Stop	This is the nucleotide number ending the region which was examined for variants (or wild-type confirmation) during testing, and is to be used in conjunction with *Region of Interest Start*.
OBX	CWE	O	0…1		69547-8	Reference Nucleotide	For genotype tests, this is the nucleotide sequence within the region of interest on the reference sequence, and is to be used in conjunction with *Variable Nucleotide*.
OBX	CWE	O	0…1		69551-0	Variable Nucleotide	For genotype tests, this is the nucleotide sequence for the variation (or wild-type confirmation) tested for, and is to be used in conjunction with *Reference Nucleotide*.
OBX	CWE	O	0…1		48010-3	Genechip ID	For Genechip based tests, this contains the Genechip identifier, and is to be used in conjunction with Genechip *Manufacturer ID* and *Genechip Version*.
OBX	CWE	O	0…1		48009-5	Genechip Manufacture r ID	For Genechip based tests, this contains the Genechip manufacturer, and is to be used in conjunction with Genechip *ID* and *Genechip Version*.

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OBX	CWE	O	0…1		48011-1	Genechip Version	For Genechip based tests, this contains the Genechip version number, and is to be used in conjunction with Genechip *Manufacturer ID* and *Genechip ID*.

7.3.3 Findings

7.3.3.0 Genetic Analysis Discrete Result Panel

The *Genetic Analysis Discrete Result Panel* is a child OBR of the Genetic Analysis Master Panel, and will contain child OBR's defining the discrete findings. Currently, these are DNA sequence variants (in *DNA Analysis Discrete Sequence Variant Panel*), but will expand in future releases of the guide to include other types of genetic biomarkers, as mentioned in Chapter 9.

TABLE 7–5 – GENETIC ANALYSIS DISCRETE RESULT PANEL

OBR/ OBX	DT	Usage	Card-inality	Value Set	LOINC Code	LOINC Element Name	Description/Comments
OBR		R	1..n	N/A	55207-5	Genetic Analysis Discrete Result Panel	This is the parent OBR for genetic panels of genetic findings (e.g. *DNA Analysis Discrete Sequence Variant Panel).*

7.3.3.1 DNA Analysis Discrete Sequence Variation Panel

The *DNA Analysis Discrete Sequence Variant Panel* corresponds to the Genetic Variation model SequenceVariation class. It describes the characteristics of an identified SequenceVariation - either of clinical relevance, or as a benign difference from the reference sequence, and reported for completeness.

TABLE 7–6 – DNA ANALYSIS DISCRETE SEQUENCE VARIATION PANEL

OBR/ OBX	OBX-2 Value Type	Usage*	Card-inality	Value Set	LOINC Code	LOINC Element Name	Description/Comments
OBR		R	1..n	N/A	55208-3	DNA Analysis Discrete Sequence Variant Panel	The set of observations representing all identified DNA Sequence Variations (variants and wild type) for all of the genetic assays performed for a single accession.
OBX	CWE	O		HGNC	48018-6	Gene Identifier	HGNC gene Identifier set by the Human Genome Organization Nomenclature Committee.

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OBX	CWE	C3		NCBI	48013-7	Genomic Reference Sequence Identifier	This field carries the ID for the genomic reference sequence. The genomic reference sequence is a contiguous stretch of chromosome DNA that spans all of the exons of the gene and includes transcribed and non transcribed stretches. For this ID use either the NCBI genomic nucleotide RefSeq IDs with their version number (see: NCBI.NLM.NIH.Gov/RefSeq), or use the LRG identifiers without transcript (t or p) extensions when they become available. (See-Report sponsored by GEN2PHEN at the European Bioinformatics Institute at Hinxton UK April 24-25, 2008). C3 = third set of conditionals. Either 'Genomic Reference Sequence Identifier' or 'Transcript Reference Sequence Identifier' is required. Both may be used.

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OBX	CWE	C3		NCBI	51958-7	Transcript Reference Sequence Identifier	This field carries the ID for the transcribed reference sequence that part of the genetic reference sequence that is converted to Messenger RNA. For this ID use either the NCBI nucleotide RefSeq IDs for transcribed DNA, plus the version number (NCBI.NLM.NIH.Gov/RefSeq), or use the LRG identifiers with transcript (t or p) extensions when they become available. (Report sponsored by GEN2PHEN at the European Bioinformatics Institute at Hinxton UK April 24-25, 2008). The NCI RefSeq transcripts IDs have a prefix of “NM” for genes from the nuclear chromosomes. NCBI does not currently provide a transcript RefSeq for mitochondrial genes. The LRG transcripts Identifiers have a prefix of “LRG_” plus a t extension. Mitochondrial genes are out of scope of LRG. C3 = third set of conditionals. Either ‘Genomic Reference Sequence Identifier’ or ‘Transcript Reference Sequence Identifier’ is required (with version number). Both may be used.
OBX	CWE	O			48008-7	Allele Name	The published, commonly used name for a gene allele is recommended if it exists.
OBX	CWE	O		NCBI	48003-8	DNA Sequence Variation Identifier	A DNA Sequence Variation identifier conveys a universal or standard repository identifier for definitive characteristics of a DNA Sequence Variation. (If available, recommend using NCBI dbSNP ids - RS#)

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OBX	CWE	C4		HGVS	48004-6	DNA Sequence Variation	Human Genome Variation Society (HGVS) nomenclature for a single or set of DNA Sequence Variation(s) identified in testing. The use of the nomenclature is also used to describe non-variations (aka. wild types). Either the DNA Sequence Variation is required or the Amino Acid Change. NOTE: If NCBI's dbSNP IDs (RS#) is used, then the DNA Sequence Variation is required to uniquely define the Variant, as the number is unique to the nucleotide location and requires the details of nucleotide change. C4 = fourth set of conditionals. Either 'DNA Sequence Variation' or 'Amino Acid Change' is required. Both may be used.
OBX	CWE	O		HGVS	48019-4	DNA Sequence Variation Type	Codified type for associated DNA Sequence Variation. DNA Sequence Variations use the HGVS notation which implies the DNA Sequence Variation Type, but the concurrent use of this code will allow a standard and explicit type for technical and display convenience. LOINC Answer List values can be seen in Table 7.8.

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OBX	CWE	C4		HGVS	48005-3	Amino Acid Change	Human Genome Variation Society (HGVS) nomenclature for an amino acid change. This value is derivable from the DNA Sequence Variation value if available. It is provided for convenience. The use of the nomenclature is also used to describe non-variations (aka. wild types). Either the DNA Sequence Variation is required or the Amino Acid Change. C4 = fourth set of conditionals. Either 'DNA Sequence Variation' or 'Amino Acid Change' is required. Both may be used.
OBX	CWE	O		HGVS	48006-1	Amino Acid Change Type	Codified type for associated Amino Acid Change. Amino Acid Change's use the HGVS notation which implies the Amino Acid Change Type, but the concurrent use of this code will allow a standard and explicit type for technical and display convenience. LOINC Answer List values can be seen in Table 7.8.
OBX	CWE	O			47999-8	DNA Region Name	A human readable name for the region of interest. Typically Exon #, Intron # or other. NOTE: This is not standardized and is mainly for convenience and display purposes.

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OBX	CWE	O			53034-5	Allelic State	The level of occurrence of a single DNA Sequence Variation within a set of chromosomes. Heterozygous indicates the DNA Sequence Variation is only present in one of the two genes contained in homologous chromosomes. Homozygous indicates the DNA Sequence Variation is present in both genes contained in homologous chromosomes. Hemizygous indicates the DNA Sequence Variation exists in the only single copy of a gene in a non-homologous chromosome (the male X and Y chromosome are non-homologous). Hemiplasmic indicates that the DNA Sequence Variation is present in some but not all of the copies of mitochondrial DNA. Homoplasmic indicates that the DNA Sequence Variation is present in all of the copies of mitochondrial DNA. LOINC Answer List values can be seen in Table 7.8.
OBX	CWE	O			48002-0	Genomic Source Class	The genomic class of the specimen being analyzed: germline for inherited genome, somatic for cancer genome, and prenatal for fetal genome. LOINC Answer List values can be seen in Table 7.8.
OBX	ST	O			47998-0	DNA Sequence Variation Display Name	Thumbnail "textual display" convention of a DNA Sequence Variation and its interpretation. For example: c.2235_2249del (Glu746_Ala750del), Exon 19, EGFR, Responsive

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If reporting a genetic test specifically performed for identification of DNA Sequence Variations associated with disease, then the *Genetic Disease Sequence Variation Interpretation* should be used for coding interpretations at the DNA Sequence Variation level.							
OBX	CWE	C5			53037-8	Genetic Disease Sequence Variation Interpre-tation	Interpretation of the pathogenicity of the DNA Sequence Variation in the context of the assessed genetic disease. LOINC Answer List values can be seen in Table 7.8.
If reporting a genetic test specifically performed for identification of DNA Sequence Variations associated with drug metabolism, then the *Drug Metabolism Sequence Variation Interpretation* should be used for coding interpretations at the DNA Sequence Variation level.							
OBX	CWE	C5			53040-2	Drug Metabolism Sequence Variation Interpre-tation	Predicted phenotype for drug efficacy. A sequence variation interpretation value known to allow (responsive) or prevent (resistant) the drug to perform. LOINC Answer List values can be seen in Table 7.8.
If reporting a genetic test specifically performed for identification of DNA Sequence Variations associated with drug efficacy, then the *Drug Efficacy Sequence Variation Interpretation* should be used for coding interpretations at the DNA Sequence Variation level.							
OBX	CWE	C5			51961-1	Drug Efficacy Sequence Variation Interpre-tation	Predicted phenotype for ability of drug to bind to intended site in order to deliver intended effect. A Sequence Variation interpretation value known to allow (responsive) or prevent (resistant) the drug to perform. LOINC Answer List values can be seen in Table 7.8.
If reporting a genetic test specifically performed for determining the presence or absence of a known set of DNA Sequence Variations (e.g. molecular profile of the tumor) without an association to phenotype, then the *Genetic Variant Assessment* should be used for coding interpretations at the DNA Sequence Variation level.							
OBX	CWE	C5			69548-6	Genetic Variant Assessmen t	Denoting the genotyping result (e.g. present, absent, or no call) for a known set of mutations within a genetic profiling test. LOINC Answer List values can be seen in Table 7.8.

*R = Required; O = Optional; C3 = Conditional, one of the C3 values is required. Both can be provided. C4 = Conditional, one of the C4 values is required. Both can be provided. C5 = Conditional. Only one of the values can be used with the following rules: C5 rules – If the test is performed for diagnostic confirmation or identification of pre-clinical disease risk, then *Genetic Disease Analysis Overall Interpretation* is used in the *Genetic Analysis Summary Panel* and *Genetic Disease Sequence Variation Interpretation* is used in *DNA Analysis Discrete Sequence Variant*

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Panel. If the test is performed for identification of carrier status, then *Genetic Disease Analysis Overall Carrier Interpretation* is used in the *Genetic Analysis Summary Panel* and *Genetic Disease Sequence Variation Interpretation* is used in *DNA Analysis Discrete Sequence Variant Panel*. If a pharmacogenomic test is performed for indication of drug efficacy, then *Drug Efficacy Analysis Overall Interpretation* is used in the *Genetic Analysis Summary Panel* and *Drug Efficacy Sequence Variation Interpretation* is used in the *DNA Analysis Discrete Sequence Variant Panel*. If a pharmacogenomic test is performed for indication of drug metabolism, then *Drug Metabolism Analysis Overall Interpretation* is used in the *Genetic Analysis Summary Panel* and *Drug Metabolism Sequence Variation Interpretation* is used in the *DNA Analysis Discrete Sequence Variant Panel*.

7.4 LOINC CODES

TABLE 7-7 – LOINC CODES

LOINC #	Component	Property	Time	System	Scale	Method
47998-0	DNA Sequence Variation display name	Txt	Pt	Bld/Tiss	Nar	Molgen
47999-8	DNA region name	ID	Pt	Bld/Tiss	Nom	Molgen
48002-0	Genomic source class (LOINC Answer List values can be seen in table 7.8)	Type	Pt	Bld/Tiss	Nom	Molgen
48003-8	DNA Sequence Variation identifier	ID	Pt	Bld/Tiss	Nom	Molgen
48004-6	DNA Sequence Variation	Find	Pt	Bld/Tiss	Nom	Molgen
48005-3	Amino acid change	Find	Pt	Bld/Tiss	Nom	Molgen
48006-1	Amino acid change type (LOINC Answer List values can be seen in table 7.8)	Type	Pt	Bld/Tiss	Nom	Molgen
48008-7	Allele name	ID	Pt	Bld/Tiss	Nom	Molgen
48013-7	Genomic reference sequence identifier	ID	Pt	Bld/Tiss	Nom	Molgen
48018-6	Gene identifier	ID	Pt	Bld/Tiss	Nom	Molgen
48019-4	DNA Sequence Variation type (LOINC Answer List values can be seen in table 7.8)	Type	Pt	Bld/Tiss	Nom	Molgen
51958-7	Transcript reference sequence identifier	ID	Pt	Bld/Tiss	Nom	Molgen
51959-5	DNA region of interest	Prod	Pt	Bld/Tiss	Nom	Molgen
51961-1	Drug efficacy sequence variation interpretation (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Ord	Molgen
51963-7	Medication assessed	Prid	Pt	Bld/Tiss	Nom	Molgen
51964-5	Drug efficacy analysis overall interpretation (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Nom	Molgen

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TABLE 7-7 - LOINC CODES						
LOINC #	**Component**	**Property**	**Time**	**System**	**Scale**	**Method**
51967-8	Genetic disease assessed	ID	Pt	Bld/Tiss	Nom	Molgen
51968-6	Genetic Disease Analysis Overall Interpretation (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Nom	Molgen
51969-4	Genetic analysis summary report	Find	Pt	Bld/Tiss	Doc	Molgen
51971-0	Drug metabolism analysis overall interpretation (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Nom	MolGen
53034-5	Allelic state	Find	Pt	Bld/Tiss	Nom	Molgen
53037-8	Genetic disease sequence variation interpretation (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Nom	Molgen
53039-4	Genetic disease analysis overall carrier interpretation (LOINC Answer List values can be seen in table 7.8)	Type	Pt	Bld/Tiss	Nom	Molgen
53040-2	Drug metabolism sequence variation interpretation (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Nom	Molgen
53577-3	Reason for study additional note	Txt	Pt	Bld/Tiss	Nar	Molgen
55208-3	DNA Analysis Discrete Sequence Variation Panel	-	Pt	Bld/Tiss	-	Molgen
55207-5	Genetic Analysis Discrete Result Panel	-	Pt	Bld/Tiss	-	Molgen
55233-1	Genetic analysis master panel	-	Pt	Bld/Tiss	-	Molgen
55232-3	Genetic analysis summary panel	-	Pt	Bld/Tiss	-	Molgen
51956-1	DNA Region Analysis Test Coverage Panel	-	Pt	Bld/Tiss	-	Molgen
53041-0	DNA Region of Interest Panel	-	Pt	Bld/Tiss	-	Molgen
69545-2	Region of Interest Start	Prid	Pt	Bld/Tiss	Nom	Molgen
69546-0	Region of Interest Stop	Prid	Pt	Bld/Tiss	Nom	Molgen
69547-8	Reference Nucleotide	Prid	Pt	Bld/Tiss	Nom	Molgen
69551-0	Variable Nucleotide	Prid	Pt	Bld/Tiss	Nom	Molgen
48010-3	Genechip ID	ID	Pt	Bld/Tiss	Nom	Molgen
48009-5	Genechip Manufacturer ID	ID	Pt	Bld/Tiss	Nom	Molgen

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TABLE 7-7 – LOINC CODES						
LOINC #	**Component**	**Property**	**Time**	**System**	**Scale**	**Method**
48011-1	Genechip Version	ID	Pt	Bld/Tiss	Nom	Molgen
69548-6	Genetic Variant Assessment (LOINC Answer List values can be seen in table 7.8)	Imp	Pt	Bld/Tiss	Nom	Molgen
69549-4	Genetic Knowledge Reference	Prid	Pt	Bld/Tiss	Nom	Molgen
69550-2	Genetic Knowledge Reference Type (LOINC Answer List values can be seen in table 7.8)	Type	Pt	Bld/Tiss	Nom	Molgen

7.5 LOINC ANSWER LISTS

TABLE 7-8 – LOINC ANSWER LISTS				
LOINC code	**LOINC component**	**Sequence**	**Answer text**	**LOINC answer code**
53034-5	Allelic state	1	Heteroplasmic	LA6703-8
		2	Homoplasmic	LA6704-6
		3	Homozygous	LA6705-3
		4	Heterozygous	LA6706-1
		5	Hemizygous	LA6707-9
48006-1	Amino acid change type	1	Wild type	LA9658-1
		2	Deletion	LA6692-3
		3	Duplication	LA6686-5
		4	Frameshift	LA6694-9
		5	Initiating Methionine	LA6695-6
		6	Insertion	LA6687-3
		7	Insertion and Deletion	LA9659-9
		8	Missense	LA6698-0
		9	Nonsense	LA6699-8
		10	Silent	LA6700-4
		11	Stop Codon Mutation	LA6701-2
48019-4	DNA sequence variation type	1	Wild type	LA9658-1
		2	Deletion	LA6692-3
		3	Duplication	LA6686-5
		4	Insertion	LA6687-3
		5	Insertion/Deletion	LA6688-1
		6	Inversion	LA6689-9
		7	Substitution	LA6690-7
51964-5	Drug efficacy analysis overall interpretation	1	Responsive	LA6677-4
		2	Resistant	LA6676-6
		3	Negative	LA6577-6
		4	Inconclusive	LA9663-1
		5	Failure	LA9664-9
51961-1	Drug efficacy sequence variation interpretation	1	Resistant	LA6676-6
		2	Responsive	LA6677-4
		3	Presumed resistant	LA9660-7

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TABLE 7-8 – LOINC ANSWER LISTS				
LOINC code	**LOINC component**	**Sequence**	**Answer text**	**LOINC answer code**
		4	Presumed responsive	LA9661-5
		5	Unknown Significance	LA6682-4
		6	Benign	LA6675-8
		7	Presumed Benign	LA6674-1
		8	Presumed non-responsive	LA9662-3
51971-0	Drug metabolism analysis overall interpretation	1	Ultrarapid metabolizer	LA10315-2
		2	Extensive metabolizer	LA10316-0
		3	Intermediate metabolizer	LA10317-8
		4	Poor metabolizer	LA9657-3
		5	Inconclusive	LA9663-1
53040-2	Drug metabolism sequence variation interpretation	1	Ultrarapid metabolizer	LA10315-2
		2	Extensive metabolizer	LA10316-0
		3	Intermediate metabolizer	LA10317-8
		4	Poor metabolizer	LA9657-3
		5	Unknown Significance	LA6682-4
53039-4	Genetic disease analysis overall carrier interpretation	1	Carrier	LA10314-5
		2	Negative	LA6577-6
		3	Inconclusive	LA9663-1
		4	Failure	LA9664-9
51968-6	Genetic disease analysis overall interpretation	1	Positive	LA6576-8
		2	Negative	LA6577-6
		3	Inconclusive	LA9663-1
		4	Failure	LA9664-9
53037-8	Genetic disease sequence variation interpretation	1	Pathogenic	LA6668-3
		2	Presumed pathogenic	LA6669-1
		3	Unknown significance	LA6682-4
		4	Benign	LA6675-8
		5	Presumed benign	LA6674-1
48002-0	Genomic source class	1	Germline	LA6683-2
		2	Somatic	LA6684-0
		3	Prenatal	LA6685-7
		4	Likely Germline	LA18194-3
		5	Likely Somatic	LA18195-0
		6	Likely Prenatal	LA18196-8
		7	Unknown Genomic Origin	LA18197-6
69548-6	Genetic Variant Assessment	1	Present	LA9633-4
		2	Absent	LA9634-2
		3	No Call	LA18198-4
69550-2	Genetic Knowledge Reference Type (LOINC Property = Type)	1	Disease	LA18199-2
		2	Drug	LA18200-8
		3	Gene	LA18201-6
		4	DNA Sequence Variant	LA18202-4
		5	Amino Acid Change	LA18203-2
		6	Test	LA7556-9

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7.6 SPECIAL SYNTAX AND NOTES ON USE OF LOINC TERMS

7.6.1 DNA Sequence Variation (see table 7-5)

The expression or nomenclature used to describe SNP locations for sequence variations is as follows:

c.-32_129+10^^HGVS

Discussions regarding the uniform and unequivocal description of sequence variants in DNA and protein sequences (mutations, polymorphisms) were initiated by two papers published in 1993; Beaudet AL & Tsui LC and Beutler E. The original suggestions presented were widely discussed, modified, extended and ultimately resulted in nomenclature recommendations that have been largely accepted and are applied world-wide.

The most important rule is that all variants should be described at the most basic level, i.e. the DNA level. Descriptions should always be in relation to a **reference sequence** (see section 6.1.5), either a **genomic** or a **coding DNA** reference sequence. Discussions on which type of reference sequence to prefer, genomic or coding DNA, have been very lively. Although theoretically a genomic reference sequence seems best, in practice a coding DNA reference sequence is preferred (*see Reference Sequence discussions*). Finally, version numbers are required to uniquely identify the reference.

http://www.hgvs.org/mutnomen/refseq.html

7.6.2 Genomic Source Class

Where testing of germline reference material may not have been performed (due to workflow or unavailability of the specimen), the Genomic Source Class list as been extended to support 'Likely Somatic', 'Likely Germline', 'Likely Prenatal' and 'Unknown Genomic Origin'.

It is strongly recommended to use the full list of *Genomic Source Class* values at the mutation level; however, top level genomics source class (around the testing context) should only use the original three (somatic, germline or prenatal) as it codifies the testing context (i.e. tumor testing, inherited genetic testing, or prenatal testing).

7.6.3 Genetic Variant Assessment

Where testing scenarios are intended to assess the presence or absence of a known set of DNA variants (e.g. tumor profiling using genotyping technology), then the Genetic Variant Assessment answer list supports structured communication of these findings. Of note, 'No Call' is different from 'Absent', because 'No Call' did not result in the determination of the marker's presence or absence. This may be due to test failure or specimen specific context which renders the test ineffective. If no result is produced, it can be for one of two reasons: 1) the assay failed or 2) the region of the chromosome/gene containing the sequence variation being genotyped is deleted. For instance, if a portion of the PTEN gene is deleted, then all assays for mutations within the deleted region would be 'no call'. In this scenario, it should not be assumed that this region is deleted, because the assays covering this region of interest may have simply failed. Sequencing remains the gold standard for identification of these variants.

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8.Example Genetic Test Laboratory Messages

Note: This chapter is based on Chapter 7 (Example Laboratory Result Messages) of the parent implementation guide entitled HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007. The guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required).

The examples in this section adhere to HL7 publishing requirements in that all persons and facilities are fictitious. They are taken from the *HL7 Version 3 Publishing Facilitator's Guide, Appendix D, Storyboard Names*.

Note: Genetic specific vocabularies are in the process of being registered with HL7. As the process is not yet complete at the time of balloting, the example messages append '99' to the system coding system identifier. This is temporary and should be removed, when the registration process is completed.

Emphasis has also been placed on demonstrating the use of standardized vocabularies together with local terminology.

8.1 MINIMAL MESSAGE WITH ACKNOWLEDGEMENT

For Information on 'Minimal Message with Acknowledgement' see Chapter 7 (Example Laboratory Result Messages) of the parent implementation guide entitled HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007. The guide can be found at http://www.hl7.org/Memonly/downloads/Standards_Messaging_V251/InteroperabilitySpecificationLabResultMessage_v251.zip (HL7 membership required). This includes Successful Receipt Message, Error on Receipt Message, and Reject Receipt Message.

8.2 CARDIOMYOPATHY GENETIC TEST RESULT MESSAGE

This test is used to identify sequence variations in genes associated with Dilated Cardiomyopathy. In this example, one pathogenic sequence variant and one benign sequence variant were identified. Therefore, the *Genetic Disease Analysis Overall Diagnostic Interpretation* is 'Positive' for the SNOMED coded disease of Dilated Cardiomyopathy (in the *Genetic Disease Assessed* value field).

Note: Genetic specific vocabularies are in the process of being registered with HL7. As the process is not yet complete at the time of balloting, the example messages append '99' to the system coding system identifier. This is temporary and should be removed, when the registration process is completed.

8.2.1 Example: Genetic Disease Analysis (e.g. Dilated Cardiomyopathy)

MSH-->As according to HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1, ORU^R01, HL7 Version 2.5.1, November, 2007.

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PID-->As according to HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1, ORU^R01, HL7 Version 2.5.1, November, 2007.

OBR|1||PM-08-J00094^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|lm_DCM-pnlB_L^Dilated Cardiomyopathy Panel B (5 genes)^99LMM-ORDER-TEST-ID||20080702000000|20080702100909||||||||234567891^Pump^Patrick^^^^^^NPI^L|||||20080703000000|||F|||||00000009^Cardiovascular^99HPCGG-GVIE-INDICATION^^^^^^Clinical Diagnosis and Family History of DCM|&Geneticist&Gene&&&&&NPI^^^^^^^HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO||||||||||||||55233-1^Genetic analysis master panel ^LN

SPM|1|||119273009&Peripheral blood&SNM3&&&&0707Intl&&Blood, Peripheral||||||||||||20080702000000

OBR|2||PM-08-J00094-1^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55232-3^Genetic analysis summary panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-J00094&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|51967-8^Genetic disease assessed^LN||399020009^DCM-Dilated Cardiomyopathy^SNM3^^^0707Intl|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48002-0^Genomic source class^LN||LA6683-2^Germline^LN|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|51968-6^Genetic disease analysis overall interpretation^LN||LA6576-8^Positive^LN|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|FT|51969-4^Genetic analysis summary report^LN ||\H\CASE\N\ - PM-08-J00094\.br\\.br\\H\PATIENT\N\ - Eve Everywoman\.br\\.br\\H\TEST PERFORMED\N\ - DCM-pnlB \.br\\.br\\H\TEST DESCRIPTION\N\ - Dilated Cardiomyopathy Panel B (5 genes)\.br\\.br\\H\INDICATION FOR TEST\N\ - Clinical Diagnosis and Family History of DCM\.br\\.br\\H\RESULTS\N\\.br\\.br\\.in+3\\H\DNA VARIANTS:\N\\.br\Heterozygous 1129C>T (R377C), Exon 6, LMNA, Presumed Pathogenic\.br\\.br\\H\ INTERPRETATION:\N\\.br\\H\Positive\N\. DNA sequencing of the coding regions and splice sites of the ACTC, LDB3, LMNA, PLN and TAZ genes revealed a heterozygous R377C variant in exon 6 of the LMNA gene (NM_170707.1). The R377C variant has been reported in the literature (Muchir 2000, Ki 2002, Kubben 2006, van Tintelen 2007). As such, this variant is highly likely to be pathogenic and therefore causative for DCM. Genetic testing of this patient's biological parents and other family members, particularly those who are affected, may help to confirm the significance of this variant. Please note that the laboratory can attempt testing on tissue specimens from deceased family members. It should be noted that the expression of DCM is the product not only of a DCM gene variant, but also of other modifier genes and environmental factors. The significance of a variant should always be interpreted in the context of the patient's clinical manifestations.\.br\\.br\\H\ RECOMMENDATION:\N\\.br\If you would like more information about the clinical manifestations of DCM variants we recommend you visit a cardiology center with expertise in the management of dilated cardiomyopathy such as the BWH Cardiovascular Genetics Center at 617-732-4837 (www.brighamandwomens.org/cvcenter/Services/genetics.asp). DCM caused by LMNA variants is inherited in an autosomal dominant manner where each first-degree relative of an individual with a DCM causing mutation has a 50% (or 1 in 2) chance of inheriting the mutation. Genetic testing is available for at-risk family members if desired. Genetic counseling is recommended for this patient and his family. For assistance in locating nearby genetic counseling services please call the laboratory at 617-768-8500 or email at LMM@partners.org.\.br\\.br\\H\

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COMMENTS:\N\\.br\Common sequence variants of unlikely clinical significance are not included in this report but are available upon request.\.br\\.br\\.in-3\\H\ TEST INFORMATION\N\\.br\\.br\\H\ BACKGROUND:\N\\.br\Dilated cardiomyopathy (DCM) is characterized by ventricular chamber enlargement and systolic dysfunction with normal left ventricular wall thickness. The estimated prevalence of DCM is 1/2,500 and about 20-35% of cases have a family history showing a predominantly autosomal mode of inheritance. Mutations in more than 20 genes have been shown to cause DCM, several of which (including MYH7, MYBPC3, TNNT2, TNNI3, TPM1 and ACTC), are also known to cause hypertrophic cardiomyopathy. Mutations in some genes cause additional abnormalities: Lamin A/C (LMNA) mutations are frequently found in DCM that occurs with progressive conduction system disease. Mutations in the Tafazzin (TAZ) gene cause Barth syndrome, an X-linked cardioskeletal myopathy in infants. In addition, mutations in several genes (including LMNA, DES, SGCD and EMD) can cause DCM in conjunction with skeletal myopathy. Genetic testing can confirm the diagnosis of DCM in patients with disease as well as identify at risk family members prior to the onset of symptoms.\.br\\.br\\H\ METHODOLOGY:\N\\.br\This test is performed by direct DNA sequencing of the coding regions and splice sites of the ACTC (NM_005159), LDB3 (NM_007078.2: Exons 5, 6 and 9 NM_001080116.1), PLN (NM_002667.2), LMNA (NM_170707.1) and TAZ (NM_000116.2) genes. This test does not detect large deletions or mutations in non-coding regions that could affect gene expression. This method is over 99.9% accurate. This test was developed and its performance characteristics determined by the Laboratory for Molecular Medicine, Harvard Partners Center for Genetics and Genomics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary.\.br\\.br\\H\ REFERENCES:\N\\.br\\.br\\H\ PRINIPAL INTERPRETER\N\ - Gene Geneticist created on 2008/07/02\.br\||||||F|20080702100909

OBX|5|ST|53577-3^Reason for study additional note^LN||Clinical Diagnosis and Family History of DCM||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBR|3||PM-08-J00094-2^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55207-5^Genetic analysis discrete result panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-J00094&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBR|4||PM-08-J00094-3^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55208-3^DNA analysis discrete sequence variation panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-J00094-2&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|48018-6^Gene identifier^LN||6636^LMNA^2.16.840.1.113883.6.281||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48013-7^Genomic reference sequence identifier^LN||NC_000001.9^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|51958-7^Transcript reference sequence identifier^LN||NM_170707.1^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|CWE|48004-6^DNA sequence variation^LN||c.1129C>T^^2.16.840.1.113883.6.282||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

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OBX|5|CWE|48019-4^DNA sequence variation type^LN||LA6690-7^Substitution^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|6|CWE|48005-3^Amino acid change^LN||p.R377C^^2.16.840.1.113883.6.282||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|7|CWE|48006-1^Amino acid change type^LN||LA6698-0^Missense^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|8|CWE|48003-8^DNA sequence variation identifier^LN||16322213^^99HPCGG-LMM-MARKER||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|9|CWE|47999-8^DNA region name^LN||^Exon 6||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|10|CWE|53034-5^Allelic state^LN||LA6706-1^Heterozygous^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|11|CWE|48002-0^Genomic source class^LN||LA6683-2^Germline^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|12|ST|47998-0^DNA sequence variation display name^LN||Heterozygous c.1129C>T (p.R377C), Exon 6, LMNA, Presumed Pathogenic||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|13|CWE|53037-8^Genetic disease sequence variation interpretation^LN||LA6669-1^Presumed pathogenic^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

****** *Second DNA sequence variation identified and listed in a DNA analysis discrete sequence variation panel*********

OBR|5||PM-08-J00094-3^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55208-3^DNA analysis discrete sequence variation panel ^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-J00094-2&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|48018-6^Gene identifier^LN||6636^LMNA^2.16.840.1.113883.6.281||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|48013-7^Genomic reference sequence identifier^LN||NC_000001.9^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

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OBX|4|CWE|51958-7^Transcript reference sequence identifier^LN||NM_170707.1^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|5|CWE|48004-6^DNA sequence variation^LN||c.1698C>T^^HGVS||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|6|CWE|48019-4^DNA sequence variation type^LN||LA6690-7^Substitution^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|7|CWE|48005-3^Amino acid change^LN||p.His566His^^HGVS||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|8|CWE|48006-1^Amino acid change type^LN||LA6700-4^Silent^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|9|CWE|48003-8^DNA sequence variation identifier^LN||12571327^^99HPCGG-LMM-MARKER||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|10|CWE|47999-8^DNA region name^LN||^Exon 10||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|11|CWE|53034-5^Allelic state^LN||LA6706-1^Heterozygous^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|12|CWE|48002-0^Genomic source class^LN||LA6683-2^Germline^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|13|ST|47998-0^DNA sequence variation display name^LN||Heterozygous c.1698C>T (p.His566His), Exon 10, LMNA, Presumed Benign||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|14|CWE|53037-8^ Genetic disease sequence variation interpretation^LN||LA6674-1^Presumed benign^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

8.3 WARFARIN METABOLISM GENETIC TEST RESULT MESSAGE

This test is used to identify alleles or haplotypes of the patient's CYP2C9 and VKORC1 genes. Although modeled below with clinical interpretations in the *Drug Metabolism Analysis Overall Interpretation* and *Drug Metabolism Sequence Variation Interpretation* values, the pilot laboratory does not report an

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interpretation, because other clinical data is needed to assess clinical dosage (e.g. weight and diet). In this example, the pilot laboratory sends structured results that are entered into a drug dosage calculator.

Note: Genetic specific vocabularies are in the process of being registered with HL7. As the process is not yet complete at the time of balloting, the example messages append ‘99’ to the system coding system identifier. This is temporary and should be removed, when the registration process is completed.

8.3.1 Example: Warfarin metabolism

MSH-->As according to HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007.

PID-->As according to HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007.

OBR|1||PM-08-X00912^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|lmp-WARF-pnlA^Warfarin Metabolism Panel^99LMM-ORDER-TEST-ID||20080702000000|20080702100909||||||||234567891^Pump^Patrick^^^^^^NPI^L|||||20080703000000|||F|||||182764009^Anticoagulant therapy^SNM3^^^^0707Intl|&Genetics&Gene&&&&&NPI^^^^^^^HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO||||||||||||||55233-1^Genetic analysis master panel ^LN

SPM|1|||119273009&Peripheral blood&SNM3&&&&0707Intl&&Blood, Peripheral||||||||||||20080702000000

OBR|2||PM-08-X00912-1^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55232-3^Genetic analysis summary panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-X00912&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|51963-7^Medication assessed^LN||11289^Warfarin^RxNorm|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48002-0^Genomic source class^LN||LA6683-2^Germline^LN-ANS|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|53038-6^Drug metabolism analysis overall interpretation^LN||LA10317-8^Intermediate metabolizer^LN-ANS|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|FT|51969-4^Genetic analysis summary report^LN ||\H\CASE\N\PM-08-X00912\.br\\.br\\H\PATIENT\N\ - Eve Everywoman\.br\\.br\\H\TEST PERFORMED\N\ - ...

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OBR|2||PM-08-X00912-2^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55207-5^Genetic analysis discrete result panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-X00912&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBR|3||PM-08-X00912-3^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55208-3^DNA analysis discrete sequence variation panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-X00912-2&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|48018-6^Gene identifier^LN|1|2623^CYP2C9^2.16.840.1.113883.6.281|||||F|20080702100909||||||||||Laboratory

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for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48013-7^Genomic reference sequence identifier^LN|1|NT_030059.12^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|51958-7^Transcript reference sequence identifier^LN|1|NM_000771.2^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|CWE|48008-7^Allele name^LN|1|^*1||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|5|CWE|48018-6^DNA sequence variation identifier^LN|1|rs1799853^^2.16.840.1.113883.6.284^582343^^99LMM-MARKER||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|6|CWE|48004-6^DNA sequence variation^LN||[c.430C, c.1075A]^^2.16.840.1.113883.6.282||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|7|CWE|48019-4^DNA sequence variation type^LN||LA9658-1^Wild type^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|10|CWE|47999-8^DNA region name^LN||||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|11|CWE|53034-5^Allelic state^LN||LA6705-3^Homozygous^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|12|CWE|48002-0^Genomic source class^LN||LA6683-2^Germline^LN|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|13|ST|47998-0^DNA sequence variation display name^LN||Homozygous CYP2C9 *1, Intermediate Metabolizer (-or- CYP2C9 *1/*1, Intermediate Metabolizer)||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|14|CWE|53040-2^Drug metabolism sequence variation interpretation^LN |1|^LA10317-8^Intermediate metabolizer^LN|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

********** *Second DNA Analysis Discrete sequence variation Panel* **********

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OBR|4||PM-08-X00912-3^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55208-3^DNA analysis discrete sequence variation panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-X00912-2&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|48018-6^Gene identifier^LN||23663^VKORC1^2.16.840.1.113883.6.281||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48013-7^Genomic reference sequence identifier^LN||NT_010393.15^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|51958-7^Transcript reference sequence identifier^LN||NM_024006.4^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|CWE|48008-7^Allele name^LN||^Hap B||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|5|CWE|48018-6^DNA sequence variation Identifier^LN||rs9934438^^2.16.840.1.113883.6.284^99938^^99LMM-MARKER||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|6|CWE|48004-6^DNA sequence variation^LN||c.-1639G^^2.16.840.1.113883.6.282||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|7|CWE|48019-4^DNA sequence variation Type^LN||LA9658-1^Wild type^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|8|CWE|47999-8^DNA region name^LN||^Promoter||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|9|CWE|53034-5^Allelic state^LN||LA6705-3^Homozygous^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|10|CWE|48002-0^Genomic source class^LN||LA6683-2^Germline^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|11|ST|47998-0^DNA sequence variation display name^LN||Homozygous VKORC1 Hap B, Intermediate Metabolizer||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|12|CWE|53040-2^Drug metabolism sequence variation interpretation^LN||LA10317-8^Intermediate metabolizer^LN||||||F|20080702100909||||||||||Laboratory for Molecular

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Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

8.4 TYROSINE KINASE INHIBITOR EFFICACY (PHARMACOGENOMIC) GENETIC TEST RESULT MESSAGE

This test is used to identify sequence variations in the EGFR gene of a lung tumor specimen. If the sequence variation is 1) isolated to the lung tumor (i.e. somatic) and 2) associated with drug efficacy in tyrosine kinase inhibitor therapy, then the *Drug Efficacy Analysis Overall Interpretation* is '*Responsive*' and clinical decision support can provide therapeutic recommendations. If the *Drug Efficacy Analysis Overall Interpretation* is *'Resistant'*, then a contraindication alert may be presented during medication order entry, alerting the clinician to genetic findings associated with drug resistance.

Note: Genetic specific vocabularies are in the process of being registered with HL7. As the process is not yet complete at the time of balloting, the example messages append '99' to the system coding system identifier. This is temporary and should be removed, when the registration process is completed.

8.4.1 Example: Tyrosine Kinase Inhibitor efficacy (Pharmacogenomic)

MSH-->As according to HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007.

PID-->As according to HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1 , ORU^R01 , HL7 Version 2.5.1 , November, 2007.

OBR|1||PM-08-Q00228^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|lm_EGFRa_L^EGFR Kinase Domain^99LMM-ORDER-TEST-ID||20080702000000|20080702100909||||||||234567891^Tumor^Trudy^^^^^^NPI^L|||||20080703000000|||F||||||69449002^Drug effect^SNM3^^^^0707Intl|&Genetics&Gene&&&&&NPI^^^^^^^HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO||||||||||||||55233-1^Genetic analysis master panel ^LN

SPM|1|||119273009&Peripheral blood&SNM3&&&&0707Intl&&Blood, Peripheral||||||||||||20080702000000

OBR|2||PM-08-X00912-1^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55232-3^Genetic analysis summary panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-X00912&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|51963-7^Medication assessed^LN||337525^Erlotinib^RxNorm|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48002-0^Genomic source class^LN||LA6684-0^Somatic^LN|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|51964-5^Drug efficacy analysis overall interpretation^LN||^LA6677-4^Responsive^LN|||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|FT|51969-4^Genetic analysis summary report^LN||\H\CASE\N\PM-08-Q00228\.br\\.br\\H\PATIENT\N\ - Eve Everywoman\.br\\.br\\H\TEST PERFORMED\N\ - ...

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.....continuation with Pharmacogenetic Document in HL7v2 Format.....

OBR|3||PM-08-Q00228-2^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55207-5^Genetic analysis discrete result panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-Q00228&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBR|4||PM-08-Q00228-3^HPCGG-LMM^2.16.840.1.113883.3.167.1^ISO|55208-3^DNA analysis discrete sequence variation panel^LN|||20080702000000||||||||||||||20080703000000|||F||||^PM-08-Q00228-2&HPCGG-LMM&2.16.840.1.113883.3.167.1&ISO

OBX|1|CWE|48018-6^Gene identifier^LN||3236^EGFR^2.16.840.1.113883.6.281||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|2|CWE|48013-7^Genomic reference sequence identifier^LN||NT_033968.5^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|3|CWE|51958-7^Transcript reference sequence identifier^LN||NM_005228.3^^2.16.840.1.113883.6.280||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|4|CWE|48003-8^DNA sequence variation Identifier^LN||16322213^^99HPCGG-LMM-MARKER||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|5|CWE|48004-6^DNA sequence variation^LN||c.2235_2249del^^2.16.840.1.113883.6.282||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|6|CWE|48019-4^DNA sequence variation type^LN||LA6692-3^Deletion^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|7|CWE|48005-3^Amino acid change^LN||p.Glu746_Ala750del^^2.16.840.1.113883.6.282||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|8|CWE|48006-1^Amino acid change type^LN||^LA6692-3Deletion^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|9|CWE|47999-8^DNA region name^LN||^Exon 19||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|10|CWE|48002-0^Genomic source class^LN||LA6684-0^Somatic^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|11|ST|47998-0^DNA sequence variation display name^LN||2235_2249del (Glu746_Ala750del), Exon 19, EGFR, Responsive||||||F|20080702100909||||||||||Laboratory for Molecular

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Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

OBX|12|CWE|51961-1^Drug efficacy sequence variation interpretation^LN |1|LA6677-4^Responsive^LN||||||F|20080702100909||||||||||Laboratory for Molecular Medicine^L^22D1005307^^^CLIA&2.16.840.1.113883.4.7&ISO|1000 Laboratory Lane^Ste. 123^Cambridge^MA^99999^USA^B

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9.Future Plans

Future versions of this implementation guide will be released in order to extend the HL7 2.5.1 genetic test reporting functionality to support additional testing paradigms. These include (but are not limited to):

1) Clinical scenarios (e.g. metastatic risk of cancer)
2) Size of genetic alteration (e.g. large genetic deletions spanning more than one gene)
3) Type of genetic alteration (e.g. copy number changes)
4) Molecular level of testing (e.g. RNA expression profiling)
5) Testing scenarios (e.g. genome sequencing)

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